ValleyWatch: mapping flood damage from space

Multimodal AI Hackathon 2026, Track B. Team: *[names to be added]*. Draft of 11 October 2026. Target length: six pages.

ValleyWatch is an educational prototype, not an operational tool.

1. What it does

Give ValleyWatch a bounding box and a flood date. From raw satellite data it answers three questions for rescue teams:

1. **Where did the flood hit?** New water and debris-covered ground, each polygon with a confidence level and the evidence behind it.
2. **What was damaged?** Buildings, road sections and bridges from pre-event OpenStreetMap, each with a confidence level.
3. **Who is cut off?** Settlements that lost their vehicle road, or every route, to the nearest town or hospital, with the broken link named.

A map dashboard shows the result, a one-page situation report is produced in English and Nepali, and a copilot answers questions from the computed data. Nothing is specific to one valley: area and date are the only inputs.

2. Data and rules

Input	Use
Sentinel-1 GRD (VV, VH)	change detection, same relative orbit before and after
Sentinel-2 L2A	vegetation loss and water, cloud-masked per pixel
Copernicus DEM 30 m	terrain correction, radar blind zones, river corridor, flow path
OpenStreetMap via ohsome, dated before the event	buildings, roads, bridges, places, health facilities

No other dataset is an input: no surface-water layer, population grid or other elevation model. Permanent water comes from the pre-event radar image and settlement size from building counts. Copernicus EMS maps (EMSR927) are read only by a separate validation module that the pipeline never imports.

3. Method

Scene picker. The Copernicus catalogue is searched around the date. The pair is chosen on one relative orbit and pass direction, soonest after the flood first. The pass before that is kept as a no-flood baseline. For the Trishuli case this gives orbit 85 ascending, 16 and 28 August 2026.

Radar change. openEO returns backscatter orthorectified with the Copernicus DEM. After a Lee speckle filter, change is the difference in dB. The threshold adapts to each area: three times the noise level measured on the pre-event pair, limited to 3 to 6 dB. Both bands dropping means new water; VV rising means debris. Pixels dark before the event are permanent water and are removed. Hysteresis follows thin channels.

Terrain. Slopes above 30 degrees, layover and shadow are excluded using the DEM and the look direction. A river corridor is derived from the DEM by flow routing: the core is ground within 20 m of height above the river, and a detection may grow from the core up to 120 m while it stays connected.

Optical check. Sentinel-2 composites before and after are masked per pixel with the scene classification band, so a mostly cloudy scene still contributes its clear gaps. Vegetation loss (green before, bare and bright after) adds debris that radar missed in gorges. Optical water only confirms radar water.

Flood-path filter. A flood runs downstream without gaps. Detections are kept only where the ground drains into a river reach downstream of confident change. Isolated patches in side valleys are dropped.

Confidence by agreement. High: radar and optical agree. Medium: optical only, or radar water with no clear optical view. Low: radar-only debris, radar disputed by optical, water seen only by the second track.

Damage. Buildings overlapping a polygon are "hit"; within 15 m, "possibly affected". A bridge is judged only by the bank at each end of its deck (a 25 m stretch, at least half covered), never by change in the river under the span.

Cut-off analysis. A road graph from pre-event OpenStreetMap, covering the box and 15 km around it. Hit sections and hit bridges are removed. Each settlement is tested at two levels, vehicle roads only and any route including footpaths, to the nearest town and the nearest hospital.

Thorough mode adds a second radar track looking from the opposite side.

4. How we check ourselves: the no-flood test

We had no ground truth while building, so we built a test that needs none. The identical rules are run on two images from BEFORE the flood. Whatever they map there is false alarm. A rule may be loosened only while this no-flood area stays within 5% of the flood-pair area. No setting was chosen from a reference-map score.

This test changed the design several times:

What the test showed	What we did
Radar debris: 4.98 km2 without a flood against 5.34 km2 with one	corridor, flood-path filter, larger minimum patch
Optical "new water": 0.37 km2 without a flood against 0.43 km2 with one (monsoon rivers widen every year)	optical water may only confirm radar water
Lowland (Kerala): more "debris" without a flood than with one	debris class switched off on plains

Final Trishuli result (35 x 52 km box, thorough mode): 3.30 km2 water and 4.78 km2 debris, with 0.33 km2 mapped on the no-flood pair (3.8%).

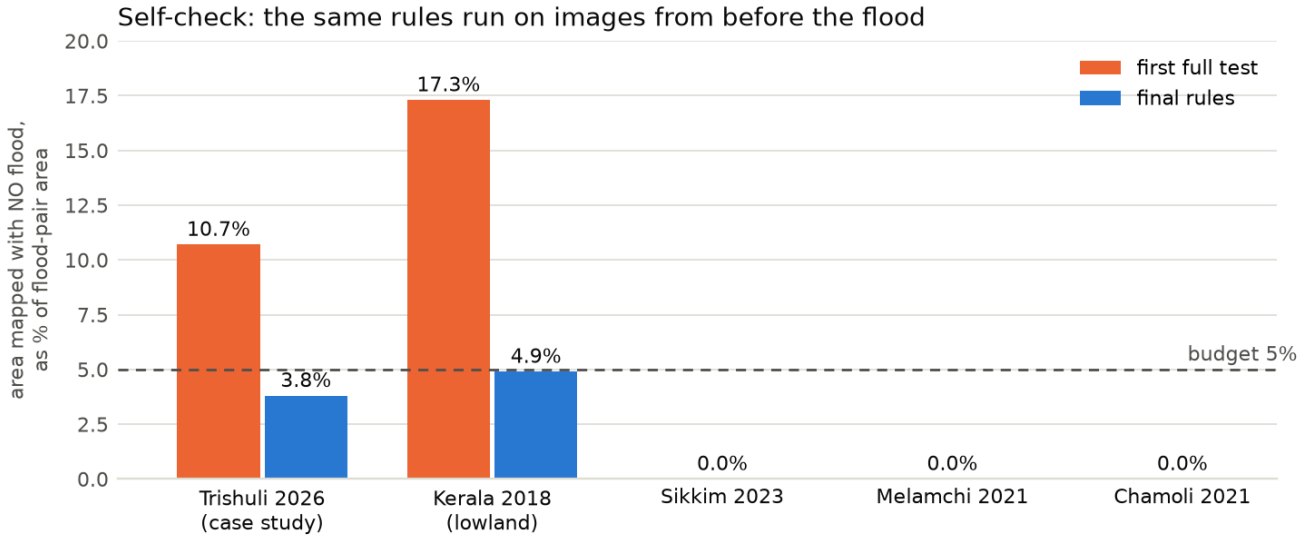


Figure 1. Area mapped when there is no flood, as a share of the flood-pair area, before and after the fixes this test led to.

5. Case study: Trishuli flood of 26 August 2026

Compared with EMSR927 (European Union, Copernicus Emergency Management Service data), which is photo-interpreted from 0.3 to 1 m images. Three of its areas lie in our box. Validation was run five times in total and never used to set a parameter.

Area	Extent recall	Extent precision	IoU	Buildings recall	Roads recall	Bridges found
Syapru Besi (gorge)	0.38	0.99	0.38	0.56	0.58	4 of 5
Timure (gorge)	0.33	0.99	0.33	0.52	0.33	0 of 1
Bidur (wide valley)	0.64	0.95	0.62	0.62	0.72	16 of 24

Thorough mode. Quick mode scores 2 to 8 points lower on extent recall.

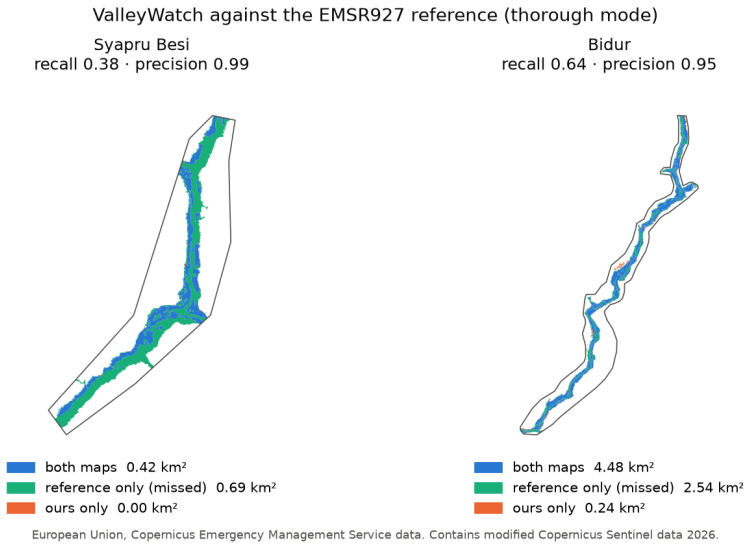


Figure 2. Our map against the reference in a gorge area (left) and the wide valley (right). European Union, Copernicus Emergency Management Service data.

What we map is almost always inside the reference area; we find about a third of it in the gorge and two thirds in the wide valley. Before the recall work (section 4) the gorge recall was 0.16.

Cut-off result for the full box: 47 of 182 settlements lost their vehicle road (about 12,800 buildings) and 9 lost every route. We have no reference for this figure. Next to it the system shows a what-if result, labelled as an assumption: the same analysis if bridges with flood change at the deck, but none on the banks, are also down.

6. Other events

Run with default settings and no changes, to see what breaks. No reference maps were used, so these show that the system runs, not that it is right.

Event	Water km2	Debris km2	No-flood share	Note
South Lhonak GLOF, Sikkim, Oct 2023	0.48	0.90	0.0%	detections follow the Teesta; no settlement points in OSM, so settlements were derived from building clusters
Melamchi, Nepal, Jun 2021	0.46	0.83	0.0%	no cut-off found
Chamoli, India, Feb 2021	0.06	0.52	0.0%	flood found at both known sites; no bridge or cut-off detected (see section 9)
Kerala lowland, Aug 2018	31.1	off	4.9%	failed the budget (17.3%) until debris was switched off on plains

7. AI component: situation-report copilot

The language model (Groq, `openai/gpt-oss-120b`) knows nothing about the event. It can only call nine functions that read the output files of a run.

It may not type a number. Each fact arrives with a placeholder such as `{buildings_hit}`; the model writes the placeholder and code fills in the value and unit. A row of a list, such as a settlement with its building count, is one placeholder, so a name and a number cannot be mixed up. If the model types a digit, a number word, or a placeholder that does not exist, the text is rejected and it gets one retry.

Asked "How many people died?" it answers that the data does not contain a death count. Asked for a percentage it would have to calculate, it says the data does not contain it.

The one-page report is built by code: every figure in its tables comes straight from the output files. The model writes only the summary paragraph, in English or Nepali, under the same rule.

8. Dashboard

One screen: map, inputs and progress on the left, totals, cut-off list, report buttons and copilot on the right. The flood map appears as soon as it is ready; damage and cut-off fill in. Clicking a total zooms to the features behind it; clicking a settlement shows its broken link and lost route. The scenes used and the last satellite pass are always visible. Bonus: click any point and the flood path runs down the valley on the DEM, listing the settlements along it.

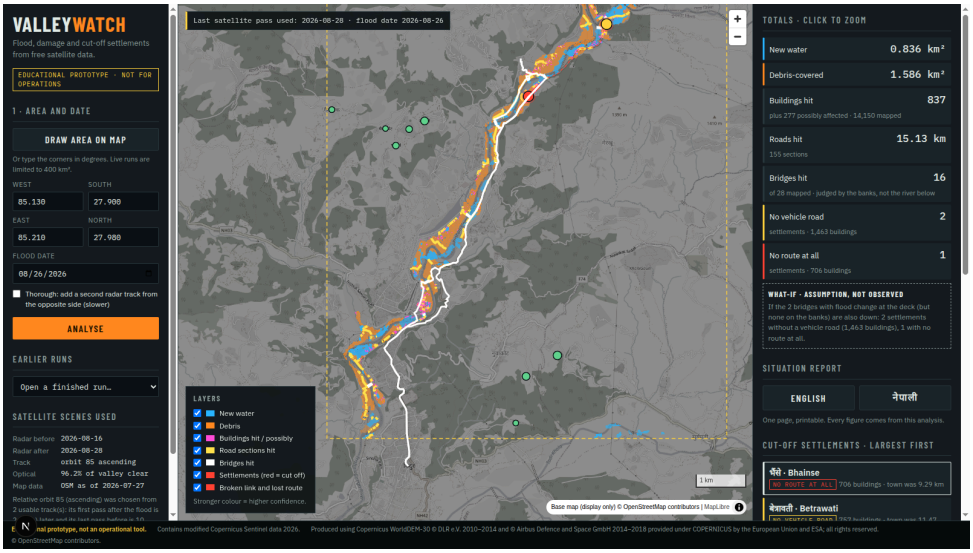


Figure 3. Dashboard after a live run near Betrawati: flood and debris, damage, a cut-off settlement with its lost route, and the what-if box.

9. Limitations

- **Not a warning system.** Sentinel-1 revisits every 6 to 12 days. The first usable image of the Trishuli flood was two days after it.
- **Gorges.** Radar cannot see layover and shadow; half of a Himalayan box can be unmappable. We find about a third of the reference area there.
- **Bridges swept away from below are missed.** We judge a bridge by its banks, so that a bridge is not flagged merely for crossing a flooded river. At Chamoli the flood stayed in the channel and the bridge at Raini, 6 m from our polygon, was reported as only "exposed". With it the cut-off analysis failed too: 0 settlements cut off, against 3 without a vehicle road and 5 without any route in the what-if result.
- **OpenStreetMap decides the damage numbers.** Kerala 2018: 28 buildings mapped in 274 km2. Chamoli 2021: half of the settlements are on no mapped road. Missing features cannot be reported as damaged.
- **Lowlands.** No debris class; the threshold reaches its ceiling where fields are noisy, so weak flooding is missed.
- **Debris and water are radar classes, not ground truth.** EMS calls the Trishuli event a mass movement; our "water" there is partly smooth mud.
- **A hit is an estimate from 10 m pixels**, not a field report. Building counts are a rough guide to people.
- **Roads outside the box are assumed intact.**
- **Second track water** can be days late; it is marked low confidence.
- **Speed and services.** A cold run of a small box takes 5 to 8 minutes and depends on three public services (Copernicus, ohsome, Groq), all of which rate-limited or failed during our tests.
- **Nepali text** in the report was written with AI help and has not been reviewed by a Nepali speaker.
- **Validation is thin.** One event with a reference map, three areas.

10. What failed, and what we learned

- A good-looking test on a small box hid a bad result on the full corridor (debris 21.8 km2). Every change is now checked on the full box and on the no-flood pair.

- The worst error was a bug: empty DEM edges were filled with high values, which dammed the river in the flow routing and turned a quarter of the box into "valley floor".
- Tile-level cloud percentages (54 to 97%) suggested Sentinel-2 was useless. Per-pixel masking found 77% of the valley floor clear.
- The first bridge rule flagged 7 of 16 bridges simply for crossing the river. The strict replacement misses bridges destroyed from below. We chose the stricter rule and report the cost.
- Fetching only main roads around the box made "no route at all" grow from 9 to 47 settlements; we kept all roads and fetched them in parallel.

Full notes with numbers are in `FINDINGS.md`.

Attribution

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AI assistance

Claude Code (Anthropic) wrote the first version of the code, the notes and this draft under the team's direction. The team set the method, the rules for self-checking and the decisions recorded in `FINDINGS.md`.